Euribor Futures Positioning Tracker

Implementation specification for Claude Code

Version 0.1 — September 2026

0. How to use this document

This is a build specification, not a research note. It is written to be handed to Claude Code as the reference for implementing a Euribor futures positioning tracker end to end.

The tool infers market positioning from publicly available open interest and price data, corrects the inference for the structural features of the Euribor complex, and anchors the resulting level to the weekly ICE Commitments of Traders report.

Build it in the order given in Section 11. Each stage produces a usable artefact and has its own acceptance tests. Do not proceed to the next stage until the current stage's tests pass.

Design principle throughout: the open interest axis is *observed*, the direction axis is *inferred*. Every design decision should preserve that asymmetry rather than collapsing it. Where the inference is weak, the code should say so explicitly through a confidence weight, not silently pick a side.

1. Scope and objectives

1.1 What the tool produces

1. A daily estimate of net speculative positioning in the Euribor futures complex, expressed in DV01 (EUR per basis point), decomposed by tenor bucket.
2. A separate curve-positioning series, capturing risk expressed through calendar spreads, packs and bundles.
3. A crowding dashboard: z-scores, percentiles, recency of accumulation, and quadrant mix.
4. A weekly reconciliation against the ICE COT report, including the fitted relationship between the daily proxy and reported category positions.

1.2 What it does not do

- It does not attempt to identify individual participants.
- It does not produce trade signals. Conditional return analysis is a diagnostic output, deliberately kept separate from the positioning estimate itself.
- It does not cover €STR futures as a primary object, though it must track their open interest for the migration correction described in Section 7.4.

1.3 Integration target

Build as a standalone module first, with its own CLI and its own tests. Once the core is stable, expose it through the existing macro-toolkit backend so it sits alongside the other tools at `localhost:8000`, following the same pattern as the existing tool files (e.g. `egb_rv.py`). Register any Bloomberg or Haver tickers it uses in the existing data catalogues rather than hard-coding them in the tool file.

Note that primary data for this tool is free ICE end-of-day and COT data, so the module must be fully functional without terminal access. Bloomberg and Haver should be optional enrichment paths, never hard dependencies.

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2. Data requirements

2.1 Primary sources

Source	Content	Frequency	Notes
ICE end-of-day files	Settlement price, volume, open interest per contract	Daily, T+1 morning	Primary. Free.
ICE Commitments of Traders	Positions by trader category, incl. spreading	Weekly	Tuesday positions, published Friday 18:30 London
ICE contract calendar	Expiry dates, first/last trading day	Static, refresh quarterly	Required for roll logic
ICE €STR futures EOD	Open interest only	Daily	For the migration correction
Optional: intraday trade data	Timestamp, price, size, aggressor flag	Tick	Materially improves direction inference
Optional: ICE spread/ strategy volume	Volume split outright vs strategy	Daily	Removes the need for spread inference

The two optional feeds change the quality of the output substantially. Request the strategy-volume breakdown early; it converts Section 6 from an inference problem into a lookup.

2.2 Canonical contract table

Every contract must have a permanent immutable identity. Generic labels are derived views, never storage keys.

```
contracts
---------
contract_id        TEXT PRIMARY KEY   -- e.g. "FEI_2027H"
root               TEXT               -- "FEI"
expiry_date        DATE
last_trading_day   DATE
contract_type      TEXT               -- "quarterly" | "serial"
listing_date       DATE
```

2.3 Daily observation table

```
daily_obs
---------
contract_id        TEXT
trade_date         DATE               -- session the data describes
knowledge_date     DATE               -- date this row became knowable
settlement_price   REAL
open_interest      INTEGER            -- end-of-day clearing snapshot
volume             INTEGER
vwap               REAL NULLABLE
open_price         REAL NULLABLE
PRIMARY KEY (contract_id, trade_date)
```

The `knowledge_date` column is mandatory and must be populated at ingestion. Open interest for session *t* is published on the morning of *t+1*. Any analysis that uses `trade_date` as the availability date is look-ahead biased, and the bias is concentrated precisely on event days where the results matter most. Every downstream query must filter on `knowledge_date`.

2.4 COT table

```
cot_obs
-------
report_date        DATE               -- Tuesday the positions describe
publication_date   DATE               -- Friday, 18:30 London
category           TEXT               -- investment firms, commercial, other financial, etc.
long_positions     INTEGER
short_positions    INTEGER
spreading          INTEGER
n_traders          INTEGER
PRIMARY KEY (report_date, category)
```

ICE computes COT open interest on a futures-equivalent basis, converting option positions using exchange delta factors: long calls and short puts become long futures-equivalent open interest, short calls and long puts become short. This matters for Section 8 — the COT measurement includes delta-hedged option risk that your raw futures-only flow series does not. Do not treat the two as measuring identical quantities without accounting for it.

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3. Repository layout

```
euribor_positioning/
├── config/
│   └── params.yaml            # all tunable parameters, no magic numbers in code
├── data/
│   ├── ingest_ice_eod.py
│   ├── ingest_ice_cot.py
│   ├── contract_calendar.py
│   └── schema.sql
├── core/
│   ├── base_signal.py         # Section 4
│   ├── quadrants.py           # Section 5
│   ├── rolls.py               # Section 6.1
│   ├── spreads.py             # Section 6.2
│   ├── dv01.py                # Section 6.3
│   └── anchor.py              # Section 8 (Kalman)
├── outputs/
│   ├── dashboard.py
│   └── diagnostics.py
├── tests/
│   ├── test_integrity.py
│   ├── test_rolls.py
│   ├── test_spreads.py
│   └── test_anchor.py
└── cli.py
```

Every tunable parameter lives in `params.yaml`. There should be no numeric constant in the core modules other than the contract multiplier.

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4. Stage one — the base signal

4.1 Rate-space convention

Euribor futures are quoted as 100 minus the rate. Store and compute everything in rate space to avoid sign errors:

```
Δr(t) = −Δp(t)
```

Apply this at ingestion, once, and never work in price space again downstream. Falling rates correspond to long futures positions.

4.2 The raw flow definition

For each contract i and session t:

```
ΔOI_i(t) = OI_i(t) − OI_i(t−1)

f_i(t) = w_i(t) · sign(Δp_i(t)) · ΔOI_i(t)
```

where `w` is the confidence weight defined below.

4.3 Direction estimator

Implement all four, selected by config, in descending order of preference. The module should report which estimator was used for each observation so that quality is auditable.

1. **Order-flow imbalance** (requires intraday with aggressor flag). Compute daily signed volume imbalance and allocate ΔOI in proportion, rather than assigning the whole day's OI change to a single side.
2. **Tick rule / Lee-Ready** on tick data where no aggressor flag exists.
3. **Close versus session VWAP.**
4. **Close versus open.**

Do not implement close-to-close as a selectable option. It embeds the overnight gap, which is cash- and news-driven rather than futures-flow-driven. In Euribor most information arrives between 08:45 and 10:00 CET on data and ECB days, so the intraday path carries genuine signal that close-to-close discards.

4.4 Confidence weighting

Rather than forcing a hard sign, weight each observation by how informative the price move is:

```
w_i(t) = min( |Δr_i(t)| / σ_i(t), cap )
```

where σ is an exponentially weighted volatility estimate of Δr for that contract, and `cap` is a config parameter (start at 2.0).

The purpose is specific. A day with large open interest creation and negligible price movement is a day on which a large passive counterparty absorbed aggressive flow. Those are simultaneously the most informative days about *size* and the least informative about *direction*. The weight prevents the classifier from asserting a direction it cannot support.

4.5 Price impact as a separate series

Compute and store, but keep out of the position estimate:

```
impact_i(t) = |Δr_i(t)| / |ΔOI_i(t)|
```

This is price impact per contract created — a measure of how much risk the market is absorbing. It compresses ahead of squeezes and is a useful standalone diagnostic. Do not let it contaminate the position level.

4.6 Turnover regime flag

```
turnover_i(t) = volume_i(t) / OI_i(t−1)
```

Classify each observation into:

- **Churn**: high volume, near-zero |ΔOI|. Day trading. Carries no positioning information — down-weight heavily.
- **Transfer**: low volume, large |ΔOI|. Typically a block or EFP. A real position changing hands — up-weight.
- **Normal**: everything else.

4.7 Accumulation with decay

Cumulating raw flow from an arbitrary start date leaves an unknown constant in the level. Apply exponential decay:

```
P_i(t) = λ · P_i(t−1) + f_i(t)

λ = exp( −ln2 / H )
```

After roughly three half-lives the initial condition has washed out. The resulting series is a well-defined estimator of *recently accumulated position* rather than a drifting integral.

Calibrate H empirically by maximising the correlation between weekly changes in P and weekly changes in the COT investment-firm net position. Do not default to 20 days. STIR positions are slower than FX or equity index positions, and the fitted H is typically longer than intuition suggests. Expose H per tenor bucket, not just globally — front-contract positions turn over faster than blues.

Until Stage 8 anchoring is in place, expose this series only as z-scores or changes. Never present it as a level.

4.8 Integrity checks (blocking)

These belong in `test_integrity.py` and must run on every ingestion:

- `|ΔOI_i(t)| ≤ volume_i(t)` for all *i*, *t*. Violations indicate vendor error or mis-stitched roll data. Fail loudly.
- Sum of per-contract ΔOI reconciles to total complex OI change reported by ICE.
- No `trade_date` observation is used where `knowledge_date > as_of_date`.
- Settlement price is present for every contract with non-zero OI.

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5. Stage two — quadrant classification

5.1 The 2×2 in rate space

Δr	ΔOI	Label	Interpretation
down	up	Long build	Fresh receiving / dovish risk added
up	up	Short build	Fresh paying / hawkish risk added
down	down	Short covering	Hawkish risk cut, often under duress
up	down	Long liquidation	Dovish risk cut

5.2 What is actually being labelled

ΔOI is the *net* change in contracts outstanding. Any single session contains new longs, new shorts, covering and liquidation simultaneously; only the residual is observable. The quadrant label therefore describes marginal net flow, not the day's activity.

Consequences for implementation:

- Single-day labels are noisy. Only expose aggregated labels at 5–10 session horizons in any user-facing output.
- Store the four quadrant series **separately**, do not collapse straight to a net figure. Build days carry information about crowding; liquidation days carry information about pain. They have different forward-return signatures and the decomposition is part of the product.

5.3 Reliability asymmetry

The two liquidation quadrants are more reliable than the two build quadrants, because closing flow is more urgent and more aggressive, so the direction assumption holds better.

Short covering has a specific and detectable signature: OI down, rates down, volume well above its 20-day average, and price impact spiking. Implement this as an explicit detector with its own confidence multiplier, applied on top of `w`.

5.4 The deadband

A fifth "unclassified" bucket is mandatory. Without it the classifier assigns quadrants to rounding noise in the back of the strip.

Two thresholds, both in `params.yaml`:

- **Price:** `|Δr| < k · σ_EW`, with k initialised at 0.3 (search 0.25–0.4). Volatility-scaled, never a fixed tick count. The same absolute move means different things in whites versus blues, and tick granularity varies across the strip.
- **Open interest:** `|ΔOI|` below a set fraction of that contract's own open interest, or below its own rolling median of `|ΔOI|`. Percentage terms, not lots.

Calibration rule: widen the deadband only while doing so improves the correlation between classified-day flow and weekly COT changes. If widening stops improving that correlation, the band is too wide.

5.5 Soft assignment

Preferred over hard labelling. Emit a probability vector over the four states per contract-day:

- The OI axis receives probability ≈ 1 on its observed sign.
- The direction axis receives a logistic in `|Δr| / σ`.

A day with large OI creation and no price move then splits its mass between long build and short build rather than being arbitrarily assigned. This is the honest representation: such a day genuinely is a large position transfer of unknown direction.

Hard labels remain available as the argmax for display purposes.

5.6 Aggregation

Classify per contract per day, then aggregate across the strip **in DV01, not lots**. Report quadrant mix as a share of total DV01 flow. The target output sentence is of the form "62% of this week's flow was short build, concentrated in reds" — a contract count does not support that statement.

Carry one further diagnostic: build days as a share of all classified days over a rolling window. High and rising indicates conviction accumulating. A flip toward liquidation dominance while the market continues to trend is the early tell that a move is running on covering rather than fresh risk.

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6. Stage three — structural corrections

6.1 Rolls

Euribor is quarterly IMM with a long listed strip plus serials. There is no single roll date: open interest drains from the expiring contract over its final weeks while every other contract simultaneously shifts one slot down the generic ladder. Two distinct problems.

6.1.1 The expiring contract. Its OI decays toward zero into EDSP. Naive signed flow reads this as a large liquidation; it is expiry. Exclude the front contract from flow calculation once its OI falls below a configured fraction of its own trailing three-month peak. Do **not** use a fixed day count — the drain profile varies with how much conviction is parked in the front.

6.1.2 Generic stitching. If a continuous "first generic" series is differenced across an expiry, ΔOI is meaningless: two different contracts are being subtracted.

> **Structural rule, non-negotiable:** never difference across a generic transition. Compute ΔOI at fixed-contract level, where each contract has a permanent identity. Only then map

> results into generic or tenor buckets for aggregation. This is the most important structural constraint in the entire build.

6.1.3 Tenor bucketing. Implement both:

- **Packs** (whites / reds / greens / blues) for desk-facing display. Note this is a step function — a contract jumps bucket overnight.
- **Time-to-expiry buckets** (0–1y, 1–2y, 2–3y, 3y+) with continuous migration, for anything being regressed or z-scored.

6.1.4 Serials. Monthly contracts carry much less open interest but sit inside the whites and cluster around near-term ECB meeting dates, making them specifically informative about policy-path positioning. Include them. Guard against double counting: a serial and the quarterly around it can hold related risk and their open interest must not be summed as though independent.

6.2 Spreads, packs and bundles

Why this correction is the hardest. ICE reports open interest leg by leg. A calendar spread creates open interest in both legs simultaneously. An uncorrected quadrant classifier then records a long build in one leg and a short build in the other — two directional positions where the market has taken one curve position.

In Euribor this is not a rounding error. Spread, pack and bundle activity is a large share of total volume. Flow tools that ignore it systematically overstate outright positioning and misread flatteners and steepeners as directional macro views.

Detection, in descending order of quality:

1. **Exchange strategy volume.** ICE identifies spread trades in the trade feed. If the outright-versus-strategy volume split can be sourced, this problem is solved by lookup rather than inference. Pursue this before building a detector.
2. **Contemporaneous ΔOI matching.** Same-session, similar-magnitude, opposite inferred direction across two contracts is a spread signature. Implement a rolling regression of each contract's ΔOI against every other's; persistently high pairwise coefficients with $|\text{ratio}|$ near one identify the structures actually being traded. Packs and bundles appear as 4-contract and 8-contract blocks with near-equal ΔOI across all legs — a distinctive fingerprint once specifically searched for.
3. **Basis projection.** Project the daily ΔOI vector onto a basis of common structures (outright level, 2s, flies, packs) and retain only the level loading as directional flow. More elegant, more fragile, and dependent on a stable basis. Implement only after (2) is working.

Validation. ICE COT breaks out a spreading category. Regress estimated spread share against reported spreading positions weekly. This is a direct test of whichever detector is built and will quickly reveal thresholds that are too loose.

Disposition. Do not discard spread-attributed flow. Route it to a second output series — curve positioning — which is valuable in its own right. The failure mode is contaminating the outright series, not tracking the spreads.

6.3 DV01 normalisation

Contract scaling. EUR 25 per basis point per contract, uniform across the strip.

That uniformity is convenient and also a trap: 10,000 lots in whites and 10,000 lots in blues carry identical DV01 and entirely different information. Whites express the policy path over the next few meetings; blues express terminal rate and structural views. Aggregating them into a single figure without a bucket breakdown destroys most of the signal.

Report at three levels:

1. Total net DV01 across the complex.
2. Net DV01 by tenor bucket.
3. Open-interest-weighted centre of gravity, in years.

The third is a single number capturing whether risk is being added at the front or the back of the strip. It frequently moves ahead of the level series and is worth surfacing independently.

Regime normalisation. Absolute DV01 is not comparable across time — total Euribor open interest has moved materially with rate volatility and with €STR migration. Compute both:

- Positioning as a share of total complex open interest (how crowded the market is internally).
- Z-score against a trailing window (how unusual it is historically).

These answer different questions and both should be exposed.

Volatility scaling. For any output intended to inform risk-taking, scale DV01 by prevailing rate volatility to obtain a risk-equivalent position. A given DV01 in a 30bp-annualised regime represents a fraction of the risk it represents at 90bp. Without this, positioning extremes cluster mechanically in high-volatility periods.

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7. Known complications

7.1 Options and delta hedging

Euribor option and midcurve open interest is large, and delta hedging of that book creates futures open interest with no directional view attached. Two options:

- Preferred: ingest option open interest, convert to delta equivalents using exchange delta factors, and net the hedging component out of futures flow.
- Minimum: flag days where option expiry or large option OI changes coincide with futures OI moves, and down-weight them.

Note this interacts with Section 2.4: ICE COT already includes options on a futures-equivalent basis, so the anchoring stage is comparing a delta-inclusive measurement to a futures-only state unless this is handled.

7.2 Block trades and EFPs

These create open interest with no corresponding order-book price formation. The turnover regime flag in Section 4.6 catches most of them. Where block data is separately available, exclude blocks from the direction estimator while retaining their OI contribution.

7.3 Settlement versus close

Use exchange settlement prices, not last trade. Settlement is what open interest is marked against and is the only consistently defined price at the back of the strip where trading is thin.

7.4 €STR migration

Directional STIR risk is split across Euribor and €STR futures, and the basis between them is itself traded. A Euribor-only DV01 series will show apparent liquidation on days that are really migration between contracts.

Track total complex DV01 across both, and treat the Euribor-only figure explicitly as a subset rather than as the market. Implement a migration flag: days where Euribor OI falls and €STR OI rises by a comparable DV01 amount should not register as liquidation.

7.5 Data revisions

ICE open interest is occasionally revised. Store observations append-only with a revision timestamp rather than overwriting, so that historical analysis can be run on either the as-published or the revised series. Backtests should use as-published.

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8. Stage four — anchoring to COT

This is the stage that converts the tool from a heuristic into an estimator with a meaningful level.

8.1 The setup

Weekly ICE COT gives a directly observed position by trader category. Daily OI-price flow gives a high-frequency but level-ambiguous signal. Combine them in a state-space model:

State equation (daily):

```
x(t) = λ · x(t−1) + β · f(t) + η(t)
```

Measurement equation (Fridays only, on the Tuesday report date):

```
z(t) = x(t) + ε(t)
```

where `x` is the latent net position of the target category (start with investment firms — hedge funds, pension funds, CTAs), `f` is the daily corrected flow from Stages one to three, and `z` is the reported COT net position.

Estimate λ, β and the noise variances by maximum likelihood over the available history.

8.2 What this buys

- A **daily nowcast** of the target category's net position that snaps back to observed truth every Friday.
- An honest **R^2** on how much of the reported position the daily proxy explains. If this number is poor, the corrections in Section 6 are not working and the problem is diagnosable rather than hidden.
- A principled **level**, so that outputs can be expressed as positions rather than only as z-scores.

8.3 Timing discipline

The COT measurement describes Tuesday positions and is published Friday at 18:30 London. The measurement update therefore enters the filter on Friday evening, not on the Tuesday it describes. Getting this wrong produces a three-day look-ahead that will look like excellent performance in backtest and will not survive live.

8.4 Category selection

Run the filter separately for each reported category. The investment-firm series is the primary output, but the commercial and other-financial series are useful as contrast — divergence between speculative and commercial positioning is itself a signal worth surfacing.

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9. Outputs

9.1 Dashboard

1. Net DV01 by tenor bucket, z-scored against 1-year and 3-year windows.
2. Crowding: percentile rank, plus the share of the current position accumulated in the last ten sessions. A stretched position built slowly and a stretched position built in a fortnight are different risks.
3. Open interest centre of gravity across the strip, in years, with a short history.
4. Quadrant mix over rolling 5- and 10-day windows, as a share of DV01 flow.
5. Curve positioning series, from the spread-attributed flow.
6. COT reconciliation panel: nowcast versus last reported, with the fitted R^2.

9.2 Diagnostics (separate from the dashboard)

- Conditional forward returns at positioning extremes, with particular attention to the window into ECB meeting dates.
- Price impact per contract created, with its own history.
- Estimator quality log: which direction estimator was used, what share of days fell into the deadband, what share was attributed to spreads.

Keep these deliberately separate from the positioning estimate. Conflating a positioning measure with a return-prediction claim is how these tools lose credibility on a desk.

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10. Validation

Beyond the blocking integrity checks in Section 4.8:

1. **Squeeze episodes.** The cumulative flow series must show sharp unwinds during known historical squeezes. Assemble a small dated list of episodes and assert against them.
2. **Event build-up.** Open interest should build into ECB dates and HICP releases and release afterwards. Test this as an average pattern in event time.
3. **COT tracking.** Weekly changes in the corrected daily proxy should correlate with weekly changes in reported COT net positions. This single number is the best summary statistic for whether the whole pipeline works, and it should improve monotonically as Stages 6.1, 6.2 and 6.3 are added. If adding a correction makes it worse, the correction is wrong.
4. **Spread share.** Estimated spread share must track the COT spreading category.
5. **Roll neutrality.** Total complex DV01 must not jump discontinuously on IMM roll dates. Assert this explicitly; it is the cleanest test that Section 6.1 works.

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11. Build order

Stage	Deliverable	Acceptance test
1	ICE EOD and COT ingestion, schema, integrity checks	All of 4.8 passes on full history
2	Contract calendar and fixed-contract ΔOI	Roll neutrality test (10.5) passes
3	Base signal with VWAP direction estimator and decay	Non-zero correlation with weekly COT changes
4	Quadrant classifier with deadband and soft assignment	Correlation improves versus stage 3
5	Spread detection and routing	Spread share tracks COT spreading (10.4)
6	DV01 normalisation and tenor bucketing	Roll neutrality holds in DV01 space
7	Kalman anchoring	Reported R^2 above an agreed threshold
8	Dashboard and diagnostics	Visual review
9	€STR migration correction, option delta adjustment	Migration flag suppresses false liquidations
10	Integration into macro-toolkit backend	Runs at localhost:8000 alongside existing tools

Stages 1 to 3 produce something usable on their own. Stages 5 and 7 are where most of the value is added. Stage 9 can be deferred but should not be dropped.

12. Summary of the ten things most likely to go wrong

1. Using `trade_date` instead of `knowledge_date`, creating a one-day look-ahead.
2. Differencing open interest across a generic contract transition.
3. Working in price space rather than rate space and inverting the sign.
4. Reading expiry drain as liquidation.
5. Recording spread activity as two opposing directional positions.
6. Aggregating lots rather than DV01 across the strip.
7. Presenting the undecayed cumulative flow as a position level.
8. Applying the COT measurement update on Tuesday rather than Friday evening.
9. Comparing a delta-inclusive COT measurement to a futures-only state without adjustment.

10. Treating Euribor open interest as the whole market while risk migrates to €STR.